



DATABASE CAPACITY · COST · OPERATING PLAN

MongoDB Atlas Flex Plan

A practical, CRM-specific assessment of monthly pricing, storage capacity, user concurrency, backups, limitations, security and the path to scale.

Prepared for
Ananttattva · CRM V2

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18 July 2026
Document class
Decision brief · v1.0

Flex in one page

5 GB

Total logical storage

Uncompressed BSON documents plus indexes. This is a hard limit.

500

Maximum connections

Database connections, not the number of CRM employees or customer records.

₹770–₹2,882

Estimated monthly database charge

Before tax/card FX charges; based on \$1 ≈ ₹96.05.

Recommendation for Ananttattva CRM V2

Use Atlas Flex for early production and controlled, low-to-moderate traffic. Plan against 4 GB, not the full 5 GB, so that indexes and growth have headroom. Target the \$8 base band while traffic is light, create alerts at 60%, 75% and 85% storage, and prepare an M10 upgrade before sustained workload or compliance requirements exceed Flex.

What is included

5 GB

Storage included

100 ops/s

Base usage band

500 ops/s

Technical throughput ceiling

Daily

Managed snapshot

Page	Contents
3	Pricing in INR and what changes the bill
4	Storage accounting and realistic CRM record capacity
5	CRM users, simultaneous usage and connection planning
6	Features, backups, security and important limitations
7	Ananttattva CRM-specific data footprint
8	Cost-control and performance operating plan
9	Upgrade triggers, implementation checklist and sources

Important:

MongoDB does not publish a guaranteed number of application users for Flex. Actual capacity depends on query design, indexes, document sizes, traffic peaks and backend connection pooling. All user figures in this report are planning estimates and must be validated with load testing.

Monthly cost in Indian rupees

Atlas Flex is pay-as-you-go and billed hourly. For 30 days of continuous use, the official price ranges from **\$8 to \$30 per month**. The amount rises with the cluster's operations-per-second usage band.

Usage band	Official monthly total	INR estimate*	Planning interpretation
0–100 ops/s	\$8	₹769	Target band for early CRM production
100–200 ops/s	\$15	₹1,441	Growing interactive usage
200–300 ops/s	\$21	₹2,017	Busy periods; review queries and caching
300–400 ops/s	\$26	₹2,497	Upgrade planning should be active
400–500 ops/s	\$30 cap	₹2,882	Flex ceiling; sustained use is an M10 signal

*Conversion uses approximately ₹96.05 per USD observed for 18 July 2026. Final invoice can differ due to billing-day exchange rate, GST/tax treatment and bank/card foreign-exchange markup.

Safe monthly budget

₹900–₹3,400

Practical cash allowance including a buffer for currency movement and possible tax/card charges. This is a budget, not a tax quotation.

Lowest-cost route

₹0 → ₹900

Stay on Free while developing; move to Flex only when backup, API access, storage or production use requires it.

What can create additional cost?

- The Flex database fee itself is capped at \$30/month for a full month of usage.
- Other Atlas services used separately—such as certain Search, Stream Processing, Data Federation, Charts or support capabilities—can appear as separate line items.
- Indian tax, bank FX markup and currency movement are outside the USD list price.

Monthly estimate = official USD usage band × invoice-day USD/INR + applicable tax/FX charges

How much CRM data fits in 5 GB?

Flex counts **dataSize** + **indexSize**. It uses logical/uncompressed BSON size, so compression must not be used to claim extra capacity. The 5 GB ceiling cannot be expanded inside Flex.

Recommended operating envelope: use only 4 GB (80%); keep 1 GB as growth, index and incident headroom.

CRM data profile	Assumed total per customer lifecycle	Approx. records within 4 GB safe budget	Suitable interpretation
Lean	12 KB	≈ 349,000 customers	Text fields, limited notes/history, small indexes, no embedded files
Standard	40 KB	≈ 104,000 customers	Lead + client + quotations + annual/compliance summary + normal indexes
Rich	150 KB	≈ 27,000 customers	Long histories, larger annual-return JSON, multiple quotations and heavier indexes

Best planning number for CRM V2: use **50,000–100,000 complete customer lifecycles** as a sensible pre-measurement range. This is deliberately below the mathematical maximum and assumes files/images are stored outside MongoDB.

Illustrative 100,000-customer standard scenario

Data family	Planning allocation	Examples
Core customer records	1.2 GB	Leads, clients, contact details, assignments
Commercial/compliance records	1.6 GB	Quotations, annual returns, approvals, PO metadata
Activities and operations	0.6 GB	Follow-ups, notifications, calendar/todo
Indexes	0.6 GB	Email, lead code, dates, status, ownership and lookups
Total target	4.0 GB	Leaves ≈1 GB emergency/growth headroom

These are sizing assumptions, not observed production measurements. After migration, use `db.stats()` and collection statistics to replace assumptions with actual average document and index sizes.

How many users can the CRM support?

Three different “user” concepts: (1) CRM application users/employees stored as documents—potentially thousands; (2) simultaneously active users—limited by workload; (3) Atlas database users—credentials used by services/admins, with a project-level limit of 100. Do not create one database credential per employee.

Workload profile	Assumption per active user	Math at 100 ops/s	Recommended simultaneous users
Light viewing	~0.125 ops/s (one DB operation every 8 sec)	≈800 theoretical	200–400
Normal CRM work	~0.5 ops/s (search, save, pagination)	≈200 theoretical	75–150
Heavy interaction/import	~2 ops/s	≈50 theoretical	20–40

Recommended bands apply conservative shared-cluster headroom and are not MongoDB guarantees. One API click can cause several database operations. Background jobs and CCP sync also consume operations.

500

Concurrent database connections

Use one pooled connection set per backend instance. A CRM employee should not open a direct Atlas connection.

500 ops/s

Hard throughput ceiling

Above the rate, Atlas queues/throttles work. The \$8 band is based on up to 100 ops/s usage.

Recommended Node/Mongoose connection policy

- Reuse a single process-level connection pool; never connect/disconnect for every API request.
- Start with a modest pool (for example 10–20 per backend instance), then tune from metrics.
- Limit serverless/backend instance fan-out so aggregate pools remain well below 500.
- Debounce search, paginate list screens and cache slow-changing dashboard aggregates.
- Run bulk imports in controlled batches outside peak hours.

Estimated DB ops/s = active users × operations per active user + sync/jobs/background traffic

What Flex provides—and what it does not

Area	Included / available	Important boundary
Storage	5 GB total logical data + indexes	Fixed; no auto-expand; hard write limit
Throughput	Usage scales through pricing bands	500 read/write operations per second ceiling
Connections	Up to 500 concurrent connections	Pool connections through the backend
Databases / collections	Up to 100 databases / 500 collections total	CRM V2 needs only a small fraction
Backup	One managed daily snapshot	No custom schedule, on-demand snapshot or point-in-time restore
Monitoring	Connections, logical size, network, opscounter alerts/metrics	No Performance Advisor or full real-time panel
Deployment	Selected AWS, Azure and GCP regions	No sharding, private endpoints, peering or dedicated nodes
Security	Atlas authentication, IP access list, TLS and managed platform security	No customer-managed encryption key or database auditing on Flex

Operational restrictions worth knowing

- Aggregation pipelines: maximum 50 stages.
- In-memory sort limit: 32 MB.
- No allowDiskUse for queries/aggregations.
- Maximum BSON nesting depth: 50.

- No configurable maintenance window.
- No downloadable database logs.
- No sharded deployment.
- Flex resources are shared, not dedicated.

Business-critical warning: A daily snapshot can allow up to approximately one day of data loss between snapshots. If the CRM requires near-zero recovery point, point-in-time restore, private networking, audit logs or customer-managed keys, choose Dedicated M10+ rather than Flex.

Ananttattva CRM V2 data footprint

The current codebase contains these principal MongoDB document families. Mixed JSON fields and growing arrays create the greatest sizing uncertainty.

Collection family	Likely size behavior	Risk / action
Users, Teams, Roles	Small and slow-growing	Index email/IDs; negligible overall storage
Leads + Lead Activities	Medium; notes and follow-up history grow	Avoid unbounded embedded history; archive old activity
Clients	Variable due to mixed data object	Measure document distribution; validate payload limits
Quotations	Medium; grows with items and terms	Keep generated PDF/files in object storage
Annual Returns	Potentially large; many mixed nested objects	Highest storage risk; split/version large sections if needed
Notifications	High record count, low value after expiry	Apply retention/TTL policy where business rules permit
Calendar / Todo	Steady operational growth	Archive completed items after defined retention
Approvals / Sync Issues	Low-to-medium audit growth	Retain required evidence; archive resolved sync logs

Do not place these binary files inside Flex

Business cards

Store image in object storage; MongoDB holds URL + metadata.

Quotation PDFs

Generate/store externally; keep immutable reference in MongoDB.

Compliance evidence

Use secure document storage; retain checksum, access policy and URL.

Storage monitoring query

```
db.stats() → monitor dataSize + indexSize
Alert thresholds: 3.0 GB (60%) · 3.75 GB (75%) · 4.25 GB (85%)
Upgrade/cleanup decision should occur before 4.5 GB (90%).
```

Capacity multiplier: keeping documents outside MongoDB is the single largest saving. A single 1 MB uploaded document consumes the same logical space as roughly 25 standard 40 KB customer lifecycles.

Keep the bill low and performance healthy

Control	Recommended implementation	Benefit
Indexes	Keep only indexes that support real query patterns: IDs, status, dates, ownership, email/phone lookup. Review duplicate/unused indexes.	Faster screens without wasting the 5 GB quota
Pagination	Use cursor-based pagination for leads, clients, notifications and activities. Never return full collections.	Fewer reads, lower network and memory use
Projection	List APIs return only columns shown; fetch full nested records only on detail pages.	Reduces data transfer and application processing
Search	Debounce 300–500 ms; require 2–3 characters; avoid leading wildcard regex on large datasets.	Prevents operation spikes
Dashboard	Cache KPI results for 30–120 seconds; avoid recalculating every widget per user.	Keeps usage near the \$8 band
Retention	Define policy for notifications, sync logs, OTPs and completed operational items; use TTL only where legally/business appropriate.	Controls long-term storage growth
Attachments	Object storage + signed URLs; MongoDB stores metadata and checksum only.	Preserves Flex storage for structured CRM data
Bulk sync	Batch CCP sync/imports, cap concurrency and schedule away from business peaks.	Avoids throttling and user-facing latency
Budgets	Create Atlas budget/billing alerts and review Summary by Service monthly.	Stops surprise spend outside Flex

Monthly operating routine

1. Record dataSize, indexSize, peak connections, average/peak opscounter and network.
2. Review top CRM endpoints by latency and database calls per request.
3. Remove only confirmed unused indexes; do not optimize blindly.
4. Forecast months-to-4.25 GB using the latest three months of growth.
5. Load test before onboarding a large team, client import or new CCP integration.

Avoid false economy: If the cluster stays in the \$21–\$30 bands, queues during peak hours, or requires stronger recovery/security, optimizing indefinitely on Flex may cost more in operational risk than upgrading to Dedicated.

When to move from Flex to Dedicated

Trigger	Action
Logical storage reaches 4.25 GB or forecast reaches it within 90 days	Archive/delete according to policy or schedule M10+ migration
Sustained >300 ops/s, recurring throttling, or Flex bill remains near \$30	Load-test and compare M10/M20 performance/cost
Peak connections repeatedly exceed 350–400	Audit pooling/fan-out; upgrade if legitimate demand remains
Need point-in-time restore or recovery point below 24 hours	Move to Dedicated with continuous cloud backup
Need private endpoints, peering, database audit or customer-managed keys	Dedicated tier required
High-traffic/performance-critical production	Consider M30+ according to MongoDB guidance and load test

Go-live checklist

DATABASE

- Nearest supported cloud region selected
- Production IP access list restricted
- Separate least-privilege DB credentials
- Connection pool configured
- Critical indexes verified

OPERATIONS

- Storage and connection alerts active
- Restore procedure documented/tested
- Attachment storage separated
- Retention policy approved
- Budget owner and upgrade owner assigned

Authoritative references

1. MongoDB Atlas Flex Costs — <https://www.mongodb.com/docs/atlas/billing/atlas-flex-costs/>
2. MongoDB Atlas Flex Limitations — <https://www.mongodb.com/docs/atlas/reference/flex-limitations/>
3. MongoDB Atlas Storage FAQ — <https://www.mongodb.com/docs/atlas/reference/faq/storage/>
4. MongoDB Atlas Service Limits — <https://www.mongodb.com/docs/atlas/reference/atlas-limits/>
5. MongoDB Atlas Billing Breakdown & Optimization — <https://www.mongodb.com/docs/atlas/billing/billing-breakdown-optimization/>
6. USD/INR reference used for illustration — XE mid-market result, approximately ₹96.05/USD, observed 18 July 2026.

Information and prices were checked on 18 July 2026 and can change. MongoDB Atlas console pricing for the selected cloud/region and the final invoice always prevail. This report is technical planning guidance, not a contractual quote, tax advice or a performance guarantee.

Prepared for Ananttattva

CRM V2 · Database planning brief